



**ISTN7MP**  
**MOBILE AND PERSONAL INFORMATION**  
**SYSTEMS**  
**HONOURS MODULE – SEMESTER 1 - 2025**

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**MOBILE AND PERSONAL INFORMATION SYSTEMS**

- PART I : MOBILE COMPUTING
- PART II : MOBILE AND PERSONAL INFORMATION SYSTEMS
- PART III : ANDROID MOBILE APP DEVELOPMENT

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## MOBILE OS – AN INTRODUCTION

- SIMILAR TO THE OS LIKE WINDOWS & LINUX THAT CONTROLS THE LAPTOPS OR PERSONAL COMPUTERS, A MOBILE OS IS A SOFTWARE THAT IS SPECIALLY DESIGNED TO CONTROL MOBILE DEVICES LIKE SMARTPHONE, TABLET, WEARABLE, GLASS, TELEVISION AND ANY OTHER HANDHELD (SMART) DEVICES.
- MOBILE OS DESIGN - THREE PHASE EVOLUTION
  - PC BASED OS
  - EMBEDDED OS
  - SMARTPHONE (MOBILE) OS

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## MOBILE OS – A REVIEW

- A MOBILE OS
  - IS A SOFTWARE PLATFORM ON TOP OF WHICH OTHER PROGRAMS CAN RUN ON THE MOBILE DEVICES
  - IS RESPONSIBLE FOR DETERMINING THE FUNCTIONS AND FEATURES OF THE MOBILE DEVICES SUCH AS THUMBWHEEL, KEYBOARD, SYNCHRONIZATION WITH APPS, EMAIL, TEXT MESSAGING, WAP, ETC.
  - WILL ALSO DETERMINE THE THIRD PARTY APPS THAT CAN ALSO BE USED IN THE DEVICE

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## POPULAR MOBILE OS – AN OVERVIEW

- **ANDROID**
  - DEVELOPED BY GOOGLE INC
  - OPEN SOURCE
- **BADA**
  - OWNED BY SAMSUNG
  - PROPRIETARY
- **BLACK BERRY OS**
  - DEVELOPED BY RESEARCH IN MOTION (RIM) FOR BLACK BERRY DEVICES
  - PROPRIETARY

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## POPULAR MOBILE OS – AN OVERVIEW

- **IPHONE OS (IOS)**
  - ORIGINALLY DEVELOPED FOR APPLE'S IPHONE DEVICES
  - BUT NOW WIDELY USED FOR IPHONE, IPOD, IPAD, ETC.
  - IOS DERIVED FROM APPLE'S MAC OS
  - PROPRIETARY
- **MEEGO OS**
  - RESULT OF MERGING TWO PRODUCTS MAEMO (NOKIA) & MOBLIN (INTEL)
  - DESIGNED TO WORK ON SMARTPHONES, NETBOOKS, TABLETS, IN-VEHICLE INFORMATION SYSTEMS, AND VARIOUS DEVICES USING INTEL ATOM AND ARMV7 ARCHITECTURES
  - ARM – ORIGINALLY, ACORN RISC MACHINE LATER ADVANCED RISC MACHINE
  - OPEN SOURCE

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## POPULAR MOBILE OS – AN OVERVIEW

- SYMBIAN OS
  - OWNED BY NOKIA
  - NOKIA DO NOT MAINTAIN SYMBIAN AS OPEN SOURCE BUT AS AN ALTERNATIVE OF OPEN AND DIRECT MODEL FOR OEMS (ORIGINAL EQUIPMENT MANUFACTURERS) AND A SMALL COMMUNITY OF PLATFORM DEVELOPMENT COLLABORATORS.
- WEB OS
  - SUCCESSOR OF PALM OS AND WAS OWNED BY PALM
  - NOW ACQUIRED AND OWNED BY HP - PROPRIETARY
  - RUNS ON LINUX KERNEL

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## POPULAR MOBILE OS – AN OVERVIEW

- WINDOWS MOBILE
  - OWNED BY MICROSOFT
  - BASED ON WINDOWS CE 5.2 KERNEL
  - PROPRIETARY

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## MOBILE OS – EVOLUTION DRIVERS

- HARDWARE
- SOFTWARE
- INTERNET

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## MOBILE OS – EVOLUTION DRIVERS - HARDWARE

- PREVIOUS MOBILE OS DO NOT SUPPORT FEATURES LIKE SENSORS (ACCELEROMETERS, GYROSCOPE, COMPASS, BAROMETER, ETC) AND TOUCH SCREENS
  - SENSORS SUCH AS ACCELEROMETERS USED TO SENSE THE ORIENTATION OF THE SCREEN
- TWO HARDWARE FACTORS TO ACHIEVE – SIZE & PROCESSING CAPABILITY
- MANUFACTURERS TARGET SMALL SIZE & HIGH PROCESSING CAPABILITY FOR MOBILE DEVICES – DIFFICULT TO ACHIEVE
- BUT FORM FACTOR WAS REDUCED ENOUGH WITH AN OPTIMAL BALANCE
- PROCESSOR CAPABILITY SCALING (FOR EX: RISC BASED PROCESSOR ARCHITECTURES)

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## MOBILE OS – EVOLUTION DRIVERS - SOFTWARE

- SOFTWARE FOR
  - LAPTOP FOCUS ON USER'S PRODUCTIVITY. FOR EXAMPLE, SUPPORT FOR KEYBOARD, MOUSE, ETC.
  - PDA (PERSONAL DATA ASSISTANT) FOCUS ON HELPING USER TO MANAGE PERSONAL DATA SUCH AS CONTACT INFORMATION, EMAIL, AND SO ON.
  - MOBILE OS IS STILL A CHALLENGE AND WERE NOT DESIGNED FOR GOOD RESPONSIVENESS OR SMOOTHNESS WITH A RICH UI.
    - MANUFACTURERS AND CONSUMERS EXPECTS MOBILE DEVICES AS A REPLACEMENT FOR LAPTOP BY REDUCING THE SIZE AND INCREASING THE PROCESSING CAPABILITY OF THE MOBILE DEVICE

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## MOBILE OS – EVOLUTION DRIVERS - INTERNET

- INTERNET DEVELOPMENT (ESPECIALLY AFTER WEB 2.0)
  - MADE PEOPLE TO LIVE IN INTERNET
  - AVAILABILITY OF ABUNDANT INFORMATION IN THE WEB TO ORGANIZE, SEARCH AND MINE
  - INVOLVING MORE AND MORE PEOPLE CONTRIBUTING TOWARDS INFORMATION, APPLICATION DEVELOPMENT AND SOCIAL INTERACTIONS
  - MOBILE OS CANNOT BE SELF CONTAINED BUT HAVE TO BE OPEN SYSTEMS

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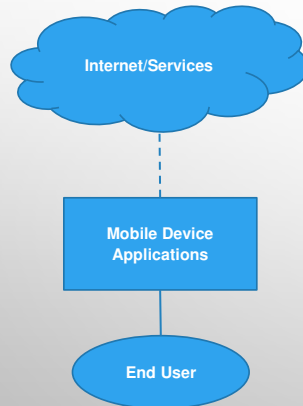
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## INTERNET USAGE MODELS

### PAST



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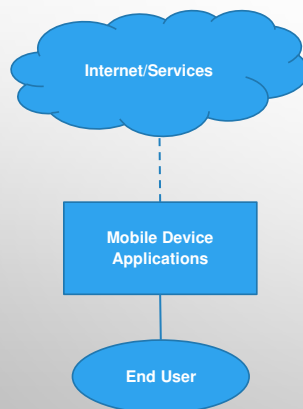
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## INTERNET USAGE MODELS

### PAST

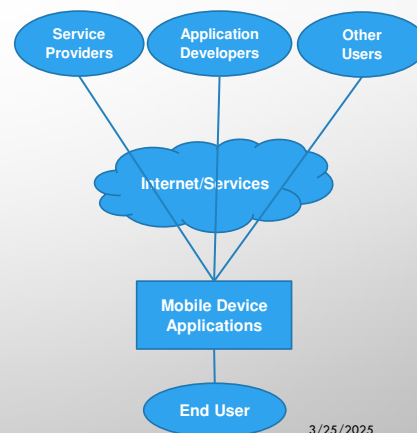


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### PRESENT



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## MOBILE OS ARCHITECTURE TRENDS

- COMMONALITIES IN MOBILE OS ARCHITECTURE

- USER EXPERIENCE
- POWER MANAGEMENT
- SECURITY DESIGN
- CLOUD SUPPORT
- OPENNESS

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## MOBILE OS ARCHITECTURE TRENDS (CONTD...)

- CONFLICTING TARGETS

- USER EXPERIENCE AND BATTERY LIFE

- TO ACHIEVE THE BEST USER EXPERIENCE, MORE SERVICES ARE AVAILABLE TO BE EXPLOITED
- AT THE SAME TIME, SUSTAINING BATTERY LIFE IS A CHALLENGE
- SOLUTION: HARDWARE COMPROMISES SHOULD BE DONE WHENEVER POSSIBLE

- SECURITY AND OPENNESS

- EXPOSING SYSTEM FUNCTIONALITIES TO EXTERNAL ENTITIES
- HENCE, OPENNESS PUTS THE SYSTEM UNDER SECURITY THREAT
- SOLUTION: STILL A CHALLENGE

- CLOUD READINESS (SOA – SERVICE ORIENTED ARCHITECTURE)

- MORE AND MORE SERVICES ARE OFFERED FROM THE CLOUD
- THIN CLIENT DEVICE MODEL MAKES IT POSSIBLE
- THIN CLIENT DEVICE TRUST CLOUD AND OFFLOADS COMPUTATION TO THE CLOUD
- THIN CLIENT MODEL STILL A CHALLENGE WITH USER EXPERIENCE AND SECURITY

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## MOBILE OS ARCHITECTURE TRENDS (CONTD...)

- CONFLICTING TARGETS

- USER EXPERIENCE AND BATTERY LIFE

- TO ACHIEVE THE BEST RESPONSIVENESS THE SYSTEM EXPECTS ALL THE RESOURCES AVAILABLE TO BE EXPLOITED
- AT THE SAME TIME, SUSTAINING BATTERY LIFE IS A CHALLENGE
- SOLUTION: HARDWARE COMPONENTS SHOULD BE IDLE WHENEVER POSSIBLE

- SECURITY AND OPENNESS

- EXPOSING SYSTEM FUNCTIONALITIES TO EXTERNAL ENTITIES
- HENCE, OPENNESS PUTS THE SYSTEM UNDER SECURITY THREAT
- SOLUTION: STILL A CHALLENGE

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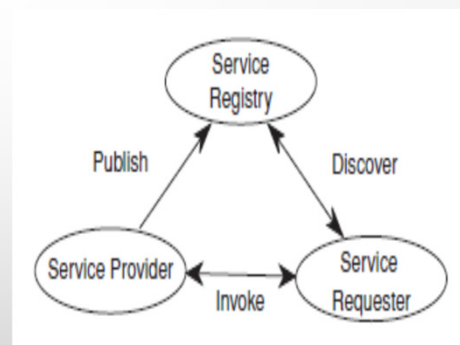
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## SERVICE ORIENTED ARCHITECTURE (SOA)

- SOFTWARE ARCHITECTURE WITH LOOSELY COUPLED SERVICES. SOA IS COMPOSED OF THREE CORE PARTS:

- THE SERVICE PROVIDER: DEFINES THE SERVICE DESCRIPTION AND PUBLISHES IT TO THE SERVICE REGISTRY
- THE SERVICE REQUESTER: ACCESSES THE REGISTER DIRECTORY TO FIND SERVICE DESCRIPTION AND ITS PROVIDERS
- THE SERVICE REGISTRY: INTERMEDIARY BETWEEN PROVIDER AND REQUESTER



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## USER EXPERIENCE

- USER EXPERIENCE IS MORE ABOUT THE DYNAMIC STATE TRANSITIONS OF THE SYSTEM TRIGGERED BY USER INPUTS
- THE QUALITY OF USER EXPERIENCE IS DETERMINED BY THE USER PERCEIVABLE RESPONSIVENESS, SMOOTHNESS, COHERENCE, AND ACCURACY
- ITS HIGH TIME TO FOCUS ON DEVICE USER INTERACTION OPTIMIZATION SO AS TO BRING THE END USER A PLEASANT EXPERIENCE
- THE TRADITIONAL PERFORMANCE OPTIMIZATION METHODOLOGY CANNOT BE SIMPLY APPLIED TO THE USER EXPERIENCE OPTIMIZATION

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## USER EXPERIENCE (CONTD...)

- USER EXPERIENCE IS MORE A SUBJECTIVE PROCESS. FOR EXAMPLE, CONSIDERING EXPERIENCE WHEN WATCHING A MOVIE OR APPRECIATING MUSIC
- CURRENT RESEARCHES FOCUS ON VARIOUS METHODOLOGIES SUCH AS EYEBALL TRACKING, HEARTBEAT MONITORING OR JUST POLLING TO UNDERSTAND USER EXPERIENCE
- SYSTEMATIC USER INTERACTIONS CAN BE CATEGORIZED INTO FOUR KINDS:
  - INPUTS TO THE DEVICE FROM THE USER, SENSOR, NETWORK AND SO ON
  - DEVICE RESPONSE TO THE INPUTS
  - SYSTEM STATE TRANSITION
  - CONTINUOUS CONTROL OF THE DEVICE

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## USER EXPERIENCE (CONTD...)

- “INPUTS TO THE DEVICE” AND “CONTROL OF THE DEVICE” ARE RELATED TO THE USER EXPERIENCE ASPECT OF **HOW A USER CONTROLS A DEVICE**
- “DEVICE RESPONSE TO THE INPUTS” AND “SYSTEM STATE TRANSITION” ARE RELATED TO THE USER EXPERIENCE ASPECT OF **HOW THE DEVICE REACTS TO THE USER**
- ANY SCENARIO CAN BE MAPPED AS A USER INTERACTION LIFE CYCLE USING THE ABOVE CATEGORIES AND IDENTIFY THE KEY METRICS IN THE SOFTWARE STACK TO MEASURE AND OPTIMISE

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## USER EXPERIENCE (CONTD...)

- USER EXPERIENCE MEASUREMENT CRITERIA FOCUS ON THE TWO COMPLIMENTARY ASPECTS
  - HOW A USER CONTROLS A DEVICE
    - ACCURACY/FUZZINESS: EVALUATES FOR ACCURACY, FUZZINESS (SOMETHING DIFFICULT TO PERCEIVE/WHEN THERE IS A RANGE OF POSSIBILITIES), RESOLUTION AND RANGE SUPPORTED BY THE SYSTEM FOR INPUTS FROM THE TOUCH SCREEN, SENSORS, AND OTHER SOURCES
    - COHERENCE: EVALUATES THE DRAG LAG DISTANCE BETWEEN THE FINGERTIP AND THE DRAGGED GRAPHIC OBJECT IN THE SCREEN. ALSO EVALUATES THE COHERENCE BETWEEN THE USER OPERATIONS AND SENSOR-CONTROLLED OBJECTS
  - HOW A DEVICE REACTS TO A USER
    - RESPONSIVENESS: EVALUATES THE TIME BETWEEN AN INPUT PROVIDED TO THE DEVICE AND DEVICE SHOWING VISIBLE RESPONSE. IT ALSO INCLUDES THE TIME SPENT TO FINISH AN ACTION
    - SMOOTHNESS: EVALUATES GRAPHIC TRANSITION SMOOTHNESS

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## USER EXPERIENCE (CONTD...)

- NO CLEAR CUT WAY AND OBJECTIVE TO MEASURE USER EXPERIENCE. THE FOLLOWING CRITERIA CAN BE CONSIDERED AS A MEASUREMENT OF THE USER INTERACTIONS:
  - PERCEIVABLE – THE METRIC NEED TO BE PERCEIVABLE BY HUMAN BEINGS
  - MEASURABLE – THE METRIC SHOULD BE MEASUREABLE BY DIFFERENT TEAMS
  - REPEATABLE – THE MEASURED RESULTS SHOULD BE REPEATABLE IN DIFFERENT MEASUREMENTS
  - COMPARABLE – THE MEASURED DATA SHOULD BE COMPARABLE ACROSS DIFFERENT SYSTEMS
  - REASONABLE – THE METRIC SHOULD HELP REASON THE CAUSALITY OF SOFTWARE STACK BEHAVIOUR
  - VERIFIABLE – THE METRIC CAN BE USED TO VERIFY AN OPTIMISATION
  - AUTOMATABLE – THE METRIC CAN BE MEASURED LARGELY ON “UNATTENDED” ASPECTS OF THE SYSTEM

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## USER EXPERIENCE (CONTD...)

- **FIVE** STEPS METHODOLOGY TO OPTIMIZE USER EXPERIENCE
  - **STEP 1:** RECEIVE THE USER EXPERIENCE **SIGHTINGS** FROM THE USERS OR IDENTIFY THE INTERACTION ISSUES WITH MANUAL OPERATIONS (USE HIGH-SPEED CAMERA OR ANY OTHER TECHNIQUE FOR OBSERVATION)
  - **STEP 2:** DEFINE THE **SOFTWARE STACK SCENARIOS AND METRICS** THAT TRANSFORM A USER EXPERIENCE INTO A SOFTWARE SYMPTOM
  - **STEP 3:** DEVELOP A SOFTWARE **WORKLOAD** (INCLUDING ALL TYPICAL USE CASES OF ANDROID DEVICE) TO REPRODUCE THE ISSUE IN A MEASURABLE AND REPEATABLE WAY. THE WORKLOAD HAS TO REPORT THE METRIC VALUES THAT REFLECT THE USER EXPERIENCE ISSUE
  - **STEP 4:** USE THE WORKLOAD AND RELATED **TOOLS** TO ANALYZE AND OPTIMIZE THE SOFTWARE STACK. THE WORKLOAD ALSO VERIFIES THE OPTIMIZATION
  - **STEP 5:** GET **FEEDBACK** FROM THE USERS AND TRY MORE APPLICATIONS WITH THE OPTIMIZATION TO CONFIRM THE USER EXPERIENCE IMPROVEMENT

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## USER EXPERIENCE (CONTD...)

- USER EXPERIENCE OPTIMIZATION IN MOBILE OS DESIGN IS MORE COMPLEX FOR FOUR REASONS:
  - INVOLVES MULTIPLE HARDWARE COMPONENTS AND MULTIPLE SOFTWARE PROCESSES AND THEIR INTERACTIONS
  - CLIENT DEVICE HAS TO CONSIDER THE POWER CONSUMPTION AS USER EXPERIENCE INCLUDES THE BATTERY LIFE AND DEVICE TEMPERATURE
  - REQUIRES PRECISE TIMING REQUIREMENTS SUCH AS SMOOTHNESS
  - SUBJECTIVE FACTORS TO BE CONSIDERED – WHETHER SOME ANIMATION IS REQUIRED OR SYSTEM CAN DROP SOME FRAMES TO GET BETTER RESPONSE FOR USER EXPERIENCE

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## POWER MANAGEMENT

- POWER MANAGEMENT IS ALWAYS A KEY CHALLENGE TO MOBILE OS DESIGNERS
- POWER DEMANDS ARE INCREASING ON MOBILE DEVICES AS MORE POWER HUNGRY APPLICATIONS ARE DEVELOPED
- BATTERY CAPACITY GROWTH IS UNABLE TO SUSTAIN THE CHALLENGE AS THE BATTERY CAPACITY DEVELOPMENT IS VERY SLOW AND DEMANDS FOR SLEEK AND COMPACT DEVICES ARE INCREASING
- 3 MAJOR FOCUS AREAS OF POWER MANAGEMENT
  - PROCESSOR POWER MANAGEMENT
  - DEVICE POWER MANAGEMENT
  - MOBILE OS POWER MANAGEMENT

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## POWER MANAGEMENT (CONTD...)

- PROCESSOR POWER MANAGEMENT
  - THE MAIN FOCUS OF MOBILE OS POWER MANAGEMENT IS ON PROCESSOR POWER MANAGEMENT AS PROCESSOR IS THE SIGNIFICANT CONSUMER OF POWER
  - MODERN PROCESSORS SUPPORT “ENHANCED INTEL SPEEDSTEP” TECHNOLOGY – THE CAPABILITY OF ADJUSTING (DYNAMIC) **PROCESSOR FREQUENCY** AND VOLTAGE SCALING AT RUN TIME
  - IN ADDITION, MODERN PROCESSORS SUPPORT MULTIPLE **PROCESSOR IDLE STATE** TO SAVE POWER - DEEPER THE IDLE STATE MORE POWER IS SAVED
  - LINUX KERNEL’S “**CPUFREQ**” AND “**CPUIDLE**” SUBSYSTEMS ARE EXAMPLES OF PROCESSOR POWER MANAGEMENT
    - AN OPERATING SYSTEM KERNEL IS THE CENTRAL AND CORE MODULE OF AN OS. IT IS THE KERNEL THAT LOADS FIRST INTO THE MEMORY AND RESIDES UNTIL THE MACHINE IS UP

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## POWER MANAGEMENT (CONTD...)

- DEVICE POWER MANAGEMENT
  - ANOTHER FOCUS OF MOBILE OS POWER MANAGEMENT IS ON DEVICE POWER MANAGEMENT
  - TECHNOLOGIES TARGET ON MECHANISMS TO MANAGE THE POWER OF I/O DEVICES AT RUNTIME
  - *RUNTIME I/O DEVICES POWER MANAGEMENT* PUT I/O DEVICES ON LOW POWER STATES WHEN THE DEVICES ARE DETECTED AS IDLE AT RUNTIME
  - IN ADDITION, I/O DEVICES ARE BECOMING SMARTER IN THE SENSE THAT THEY CAN WORK ON THEIR OWN WITHOUT CPU INTERVENTION
  - AS SEEN PREVIOUSLY WITH PROCESSORS, GPUS (GRAPHICAL PROCESSING UNITS) CAPABILITY OF VOLTAGE AND FREQUENCY SCALING REDUCE POWER CONSUMPTIONS

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## POWER MANAGEMENT (CONTD...)

- MOBILE OS POWER MANAGEMENT CASES
  - EVEN BEFORE INTRODUCING POWER MANAGEMENT INTO THE LINUX KERNEL, **ANDROID** CAME UP WITH AN APPROACH CALLED **OPPORTUNISTIC SUSPEND** TO EXTEND THE BATTERY LIFE BY SUSPENDING THE SYSTEM AGGRESSIVELY WHENEVER THERE IS NO INTERESTING WORK GOING ON (EVEN WITHOUT HOLDING A **WAKELOCK**). **WAKELOCKS ARE POWER MANAGING SOFTWARE MECHANISMS TO MAKE SURE THAT ANDROID DEVICES DO NOT GO INTO DEEP SLEEP FOR A GIVEN APP TO USE THE SYSTEM RESOURCES**
  - **WINDOWS 8** INTRODUCED A NEW SYSTEM POWER STATE CALLED **CONNECTED STANDBY** WHICH HALTS ALL SYSTEM ACTIVITIES BUT THE SYSTEM STILL RUNNING AND ENABLING USERS TO STAY UP TO DATE WITH LATEST INFORMATION SUCH AS EMAILS – CONNECTED STANDBY IS BASED ON PROCESSOR IDLE AND DEVICE RUNTIME POWER MANAGEMENT TECHNOLOGIES

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## POWER MANAGEMENT (CONTD...)

- MOBILE OS POWER MANAGEMENT CASES (CONTD...)
  - SOFTWARE HYGIENE (MAINTAINING SOFTWARE AND PREVENTING SOFTWARE PROBLEMS) IS THE MOST CHALLENGING PROBLEM FOR BOTH SUSPEND- AND IDLE- BASED POWER MANAGEMENT APPROACHES – BATTERY LIFE DEPENDS ON APPLICATION BEHAVIOURS ON SUCH SYSTEMS
  - RECENT STUDY SAYS THAT FREE ANDROID APPLICATIONS WASTE 75% OF POWER ON ADS BY HOLDING A WAKELOCK IN THE BACKGROUND, THUS BLOCK SYSTEM SUSPEND
  - THIS IS ALSO TRUE FOR WINDOWS 8 WHERE ONE POWER UNFRIENDLY APPLICATION STAYING BUSY FOR NO REASON WILL PREVENT THE ENTIRE SYSTEM FROM ENTERING CONNECTED STANDBY POWER STATE

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## OPENNESS

- ANOTHER MAJOR TREND OF MOBILE OS
- OPENNESS OF MOBILE OS REFERS TO THE LEVEL OF OPPORTUNITY AND FREEDOM THAT PEOPLE HAVE TO USE, CONTRIBUTE TO, CUSTOMIZE, AND INNOVATE FOR THE MOBILE OS FOR THEIR PURPOSES
- OPENNESS CAN BE STUDIED FROM THE DEVELOPER'S PERSPECTIVE AND ECOSYSTEM (COMMUNITY OF INTERACTING ENTITIES AND THEIR PHYSICAL ENVIRONMENT) PERSPECTIVE
- LETS FOCUS OPENNESS FROM THE ECOSYSTEM PERSPECTIVE TO REALIZE DEVELOPER'S PERSPECTIVE

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## OPENNESS (CONTD...)

- OPENNESS TO PLAYERS OF THE MOBILE ECOSYSTEM
  - THE PLAYERS OF THE MOBILE ECOSYSTEM
    - ORIGINAL EQUIPMENT MANUFACTURERS (OEM)
    - SERVICE PROVIDERS (OPERATORS)
    - CONSUMERS (END USERS)
    - INDEPENDENT SOFTWARE VENDOR (ISV)
    - DEVELOPERS
  - THE OPENNESS IMPLIES DIFFERENT THINGS TO DIFFERENT PLAYERS
  - OPENNESS IS PROVIDING ACCESS TO SOURCE CODE IN ADDITION TO APIS AND SDKS
  - EVENTUALLY THE END USERS OF MOBILE DEVICES BENEFIT FROM THE LEVEL OF OPENNESS AS THEY WILL HAVE MORE CHOICES OF APPLICATIONS AND DEVICES
  - ANDROID OS IS A GREAT EXAMPLE OF AN OPEN SOURCE MOBILE OS. THE SUCCESS OF ANDROID DEVICES INDICATES THE LEVEL OF OPENNESS

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## OPENNESS (CONTD...)

- OPENNESS TO PLAYERS OF THE MOBILE ECOSYSTEM (CONTD...)
  - THE OPENNESS IMPLIES DIFFERENT THINGS TO DIFFERENT PLAYERS
    - FOR OPERATORS THE OPENNESS OF THE MOBILE OS DETERMINES HOW EASY THEIR SERVICES CAN BE PORTED, MIGRATED, DEPLOYED, AND RUN SMOOTHLY ACROSS THE DEVICES
    - FOR OEMS, THE OPENNESS DETERMINES HOW MUCH THEY CAN CUSTOMIZE THE MOBILE OS ITSELF TO RUN ACROSS PLATFORMS AND DETERMINES HOW EASILY THEY CAN BUILD DEVICES WITH A CONSISTENT USER EXPERIENCE
    - FOR ISVS AND COMMUNITY DEVELOPERS, THE OPENNESS DETERMINES HOW EASILY THEY CAN DEVELOP NEW APPLICATIONS WITH THEIR CREATIVE IDEAS AND HOW THEIR INVESTMENT ON APPLICATION DEVELOPMENT CAN BE MAXIMIZED THROUGH PROGRAMMING ONCE RUNNING ACROSS DEVICES
    - FOR THE END USERS OR CONSUMERS OF MOBILE DEVICES, THE OPENNESS MEANS HOW EASILY THEY CAN GET MORE APPLICATIONS, LIKE THE RICH APPLICATION DOWNLOADED FROM THE STORES, WITHOUT WORRYING TOO MUCH ABOUT INCONSISTENCY OF USER EXPERIENCE AND INCOMPATIBILITY OF APPLICATION ACROSS DEVICES

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## CLOUD READINESS

- CLOUD IS BEING WIDELY USED BY MOBILE USERS
- CLOUD SERVICES ARE PRESENTED BY WEBSITES
- MORE AND MORE CLOUD SERVICES ARE PROVIDED THROUGH WEB APPLICATIONS, WHICH ARE INSTALLED FROM AN APPLICATION STORE AND RUN LIKE NATIVE APPLICATIONS ON THE MOBILE CLIENT
- WHETHER WITH A BROWSER OR STANDALONE WEB APPLICATION, THE FOLLOWING AREAS SHOULD BE CONSIDERED IN MOBILE OS DESIGN

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## CLOUD READINESS (CONTD...)

- HTML 5 CAPABILITY

- FIFTH AND CURRENT VERSION OF HTML. RELEASED DURING JANUARY 2008.
- HTML 5 SUPPORT APIS FOR COMPLEX WEB APPLICATIONS TO INTEGRATE CLOUD SERVICES WELL AND TO PROVIDE A GOOD USER EXPERIENCE
- HTML 5 IS A CANDIDATE FOR CROSS PLATFORM MOBILE APPLICATIONS
- THE MOBILE OS VENDORS VIEW HTML 5 AS MORE AND MORE IMPORTANT AND ARE MAKING IT A CORE COMPETENCY
- THE BROWSER VENDORS ARE ALSO LOOKING FOR THE POSSIBILITY TO MAKE IT DEFAULT IN THE MOBILE OS

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## CLOUD READINESS (CONTD...)

- CROSS-PLATFORM CAPABILITY

- CROSS-PLATFORM APP: AN APPLICATION THAT REACHES YOUR CUSTOMER WHATEVER THE DEVICES THEY HAVE IN FRONT OF THEM. FOR EXAMPLE, ANDROID PHONE, WINDOWS PC OR APPLE IPAD.
- HTML 5 IS WELL KNOWN FOR ITS CROSS-PLATFORM CAPABILITY
- DIFFERENT MOBILE OS PROVIDE DIFFERENT HTML 5 SUPPORT AND STANDARDIZATION OF HTML 5 IS STILL ONGOING (REFERENCE LINK: [HTTP://MOBILEHTML5.ORG/](http://mobilehtml5.org/))
- PHONEGAP IS A OPEN SOURCE CROSS PLATFORM MOBILE APPLICATION DEVELOPMENT FRAMEWORK WHICH ALLOWS MOBILE APP DEVELOPERS TO USE HTML5, CSS AND JAVASCRIPT

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## CLOUD READINESS (CONTD...)

- WEB APPLICATIONS

- WEB APPLICATIONS DEFINES THE CLIENT SIDE APPLICATIONS DEVELOPED WITH WEB TECHNOLOGIES
- WEB APPLICATIONS CAN BE INSTALLED AND RUN EVEN OFFLINE ON THE DEVICES
- WEB APPLICATIONS CAN ACCESS LOCAL DEVICES AND RESOURCES AS A NATIVE APPLICATION
- WEB APPLICATIONS IS MORE THAN A URL ACCESSED FROM A BROWSER
- TO ENABLE WEB APPLICATIONS, THE MOBILE OS NEEDS TO PROVIDE A WEB APPLICATION PLATFORM, WHICH INCLUDES WEB RUNTIME (DERIVED FROM BROWSERS), WEB FRAMEWORK (RICH LIBRARIES) AND DEVELOPMENT TOOLS

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## CLOUD READINESS (CONTD...)

- PERFORMANCE

- THE PERFORMANCE IS COMPLAINED MOST BY MOBILE APPLICATION DEVELOPERS WHEN THEY START BUILDING APPS WITH HTML5
- THE MOST IMPORTANT FOCUS AREAS TO ACHIEVE OPTIMIZATION FOR MOBILE OS
  - HARDWARE ACCELERATION
  - MULTITHREADING SUPPORT
  - JAVASCRIPT ENGINE OPTIMIZATION
  - NATIVE OR HYBRID APPLICATION SUPPORT

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## CLOUD READINESS (CONTD...)

- CLOUD INTEGRATION
  - CLOUD STORAGE SEAMLESS INTEGRATION: USAGE OF CLOUD STORAGE JUST AS THE NATIVE STORAGE BY THE CLIENT. FOR EXAMPLE, APPLE'S ICLOUD FOR IOS AND GOOGLE'S GOOGLE DRIVE FOR ANDROID
  - CLOUD APIS ACCESSIBILITY: THE MOBILE OPERATING SYSTEM SHOULD PROVIDE CAPABLE AND EASY-TO-USE LIBRARIES FOR BOTH WEB AND NATIVE APPLICATIONS TO ACCESS CLOUD CLIENT APIS
  - ACCOUNT MANAGEMENT: MULTIPLE CLIENTS WITH A SINGLE ACCOUNT SHARE THE CLOUD AND THE SYNCHRONIZATION SHOULD BE MANAGED. ACCOUNTS SHOULD BE TIGHTLY INTEGRATED IN THE MOBILE OS AND ACCOUNT APIS

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## DISCUSSION ON COMMONALITIES – APPLE IOS

- USER EXPERIENCE: IOS PROVIDES GOOD PERFORMANCE AND IS NORMALLY SET AS THE BENCHMARK FOR OTHER MOBILE OS. CONTINUOUS ENHANCEMENT OF USER EXPERIENCE PERFORMANCE
- POWER MANAGEMENT: SEEMS NOT ABLE TO CATCH UP WITH THE INCREASING DEMANDS. FOR EXAMPLE, IPHONE 4S CONSUMES TWICE AS MUCH AS DATA AS THE PREVIOUS IPHONE MODEL
- OPENNESS: IOS IS PERCEIVED AS A CLOSED MOBILE OS
- CLOUD READINESS: IOS 5.0 WITH HTML5 SUPPORT MAKES IT A GOOD CLOUD CLIENT AND THE ICLOUD HAS BEEN INTEGRATED BY DEFAULT AS STORAGE

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## DISCUSSION ON COMMONALITIES – GOOGLE ANDROID

- USER EXPERIENCE: DEFINED A SET OF DESIGN PRINCIPLES WITH THREE OVERARCHING GOALS – “ENCHANT ME”, “SIMPLIFY MY LIFE”, AND “MAKE ME AMAZING”.
- POWER MANAGEMENT: AGGRESSIVELY SUSPENDS DEVICES TO SAVE POWER WHENEVER NOTHING BLOCKS SUSPEND BY HOLDING A WAKELOCK. THIRD PARTY APPLICATIONS RUNNING IN THE BACKGROUND SUCKS POWER QUIETLY
- OPENNESS: GOOGLE RELEASES THE ANDROID CODE AS OPEN SOURCE UNDER APACHE LICENSE AND THE ANDROID OPEN SOURCE PROJECT
- CLOUD READINESS: GOOGLE DRIVE, BUT DOES NOT HAVE A COMPREHENSIVE SOLUTION AS APPLE DOES

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## DISCUSSION ON COMMONALITIES – OTHER OS

- EXPLORE

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